

Запуск ядра с init файлом

1. Очистка от старых записей:

```
tair@tair-Bravo-15-C7VE:~/dev/linux$ make mrproper
CLEAN certs
CLEAN drivers/firmware/efi/libstub
CLEAN drivers/scsi
CLEAN drivers/tty/vt
CLEAN init
CLEAN lib/crc
CLEAN lib
CLEAN usr
CLEAN .
CLEAN vmlinux.symvers modules.builtin modules.builtin.modinfo .vmlinux.export.c
CLEAN scripts/basic
CLEAN scripts/dtc
CLEAN scripts/kconfig
CLEAN scripts/mod
CLEAN scripts
CLEAN include/config include/generated .config .config.old .version
tair@tair-Bravo-15-C7VE:~/dev/linux$
```

2. Создание дефолтного конфига:

```
tair@tair-Bravo-15-C7VE:~/dev/linux$ ARCH=arm make defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
HOSTCC scripts/kconfig/expr.o
LEX scripts/kconfig/lexer.lex.c
YACC scripts/kconfig/parser.tab.[ch]
HOSTCC scripts/kconfig/lexer.lex.o
HOSTCC scripts/kconfig/menu.o
HOSTCC scripts/kconfig/parser.tab.o
HOSTCC scripts/kconfig/preprocess.o
HOSTCC scripts/kconfig/symbol.o
HOSTCC scripts/kconfig/util.o
HOSTLD scripts/kconfig/conf
*** Default configuration is based on 'multi_v7_defconfig'
#
# configuration written to .config
#
```

3. Сборка ядра (результат):

```
LD      vmlinux.o
OBJCOPY modules.builtin.modinfo
GEN      modules.builtin
MODPOST vmlinux.symvers
CC       .vmlinux.export.o
UPD      include/generated/utsversion.h
CC       init/version-timestamp.o
KSYMS    .tmp_vmlinux0.kallsyms.S
AS       .tmp_vmlinux0.kallsyms.o
LD       .tmp_vmlinux1
NM       .tmp_vmlinux1.syms
KSYMS    .tmp_vmlinux1.kallsyms.S
AS       .tmp_vmlinux1.kallsyms.o
LD       .tmp_vmlinux2
NM       .tmp_vmlinux2.syms
KSYMS    .tmp_vmlinux2.kallsyms.S
AS       .tmp_vmlinux2.kallsyms.o
LD       vmlinux
NM       System.map
SORTTAB vmlinux
OBJCOPY arch/arm/boot/Image
Kernel: arch/arm/boot/Image is ready
LDS      arch/arm/boot/compressed/vmlinux.lds
AS       arch/arm/boot/compressed/head.o
GZIP     arch/arm/boot/compressed/piggy_data
CC       arch/arm/boot/compressed/misc.o
CC       arch/arm/boot/compressed/decompress.o
CC       arch/arm/boot/compressed/string.o
AS       arch/arm/boot/compressed/hyp-stub.o
CC       arch/arm/boot/compressed/fdt_rw.o
CC       arch/arm/boot/compressed/fdt_ro.o
CC       arch/arm/boot/compressed/fdt_wip.o
CC       arch/arm/boot/compressed/fdt.o
CC       arch/arm/boot/compressed/atags_to_fdt.o
CC       arch/arm/boot/compressed/fdt_check_mem_start.o
AS       arch/arm/boot/compressed/lib1funcs.o
AS       arch/arm/boot/compressed/ashldi3.o
AS       arch/arm/boot/compressed/bswapsdi2.o
AS       arch/arm/boot/compressed/piggy.o
LD       arch/arm/boot/compressed/vmlinux
OBJCOPY arch/arm/boot/zImage
Kernel: arch/arm/boot/zImage is ready
tair@tair-Bravo-15-C7VE:~/dev/linux$
```


5. Компилирование файла init и его сжатие:

init.c

```
#include <stdio.h>
#include <unistd.h>

int main(void)
{
    printf("Hello, Tair!!!!!!!!!!!!!!!!\n");
    sleep(20);
}
```

gcc -static init.c -o init

и echo init | cpio -o -H newc | gzip > test.cpio.gz

6. Запуск ядра с файлом init и dtb

QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel
zImage -initrd test.cpio.gz -dtb vexpress-v2p-ca9.dtb -append
"console=ttyAMA0" -nographic

Результат:

```

[ 1.368432] hw perfevents: enabled with armv7_cortex_a9 PMU driver, 7 (0000003f) counters available
[ 1.371281] NET: Registered PF_INET6 protocol family
[ 1.392314] input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bus@40000000/motherboard-bus@40000000/lofpga07,00000000/10006000.kmi/serio
0/input/input0
[ 1.395636] Segment Routing with IPv6
[ 1.395792] In-situ OAM (IOAM) with IPv6
[ 1.396089] sit: IPv6, IPv4 and MPLS over IPv4 tunneling driver
[ 1.397969] NET: Registered PF_PACKET protocol family
[ 1.397997] can: controller area network core
[ 1.398228] NET: Registered PF_CAN protocol family
[ 1.398267] can: raw protocol
[ 1.398355] can: broadcast manager protocol
[ 1.398443] can: netlink gateway - max_hops=1
[ 1.399320] Key type dns_resolver registered
[ 1.399691] ThumbEE CPU extension supported.
[ 1.399740] Registering SWP/SWPB emulation handler
[ 1.435233] Loading compiled-in X.509 certificates
[ 1.448468] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
[ 1.451199] printk: console [ttyAMA0] enabled
[ 1.484019] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
[ 1.530295] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
[ 1.531772] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[ 1.841728] input: InEXPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000/motherboard-bus@40000000/lofpga07,00000000/10007000
.kmi/serio1/input/input2
[ 2.623616] clk: Disabling unused clocks
[ 2.623861] PM: genpd: Disabling unused power domains
[ 2.661081] Freeing unused kernel image (initmem) memory: 2048K
[ 2.665639] Run /init as init process
Hello, Tair!!!!!!!!!!!!
[ 12.633464] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 12.633542] amba 1000e000.memory-controller: deferred probe pending: (reason unknown)
[ 12.633556] amba 1000e1000.memory-controller: deferred probe pending: (reason unknown)
[ 12.633567] amba 1000e5000.watchdog: deferred probe pending: (reason unknown)
[ 22.741293] Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000
[ 22.742147] CPU: 0 UID: 0 PID: 1 Comm: init Not tainted 6.17.0 #1 NONE
[ 22.742451] Hardware name: ARM-Versatile Express
[ 22.742683] Call trace:
[ 22.743023] unwind_backtrace from show_stack+0x10/0x14
[ 22.743689] show_stack from dump_stack_lvl+0x54/0x68
[ 22.743863] dump_stack_lvl from vpanic+0xf4/0x304
[ 22.744015] vpanic from __do_trace_suspend_resume+0x0/0x4c
[ 22.744640] ---[ end Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000 ]---
```


Запуск ядра на базе корневой файловой системы busybox

1. Сборка дефолтного конфига:

ARCH=arm make menuconfig

```
tair@tair-Bravo-15-C7VE: ~/dev/kernel/busybox/busybox-1.38.0 190x45
command builtin (HUSH_COMMAND) [Y/n] (NEW) y
trap builtin (HUSH_TRAP) [Y/n] (NEW) y
type builtin (HUSH_TYPE) [Y/n] (NEW) y
times builtin (HUSH_TIMES) [Y/n] (NEW) y
read builtin (HUSH_READ) [Y/n] (NEW) y
set builtin (HUSH_SET) [Y/n] (NEW) y
unset builtin (HUSH_UNSET) [Y/n] (NEW) y
ulimit builtin (HUSH_ULIMIT) [Y/n] (NEW) y
umask builtin (HUSH_UMASK) [Y/n] (NEW) y
getopts builtin (HUSH_GETOPTS) [Y/n] (NEW) y
memleak builtin (debugging) (HUSH_MEMLEAK) [N/y] (NEW) n
*
* Options common to all shells
*
POSIX math support (FEATURE_SH_MATH) [Y/n/?] (NEW) y
Extend POSIX math support to 64 bit (FEATURE_SH_MATH_64) [Y/n/?] (NEW) y
Support BASE#nnnn literals (FEATURE_SH_MATH_BASE) [Y/n] (NEW) y
Hide message on interactive shell startup (FEATURE_SH_EXTRA_QUIET) [Y/n/?] (NEW) y
Standalone shell (FEATURE_SH_STANDALONE) [N/y/?] (NEW) n
Run 'nofork' applets directly (FEATURE_SH_NOFORK) [N/y/?] (NEW) n
read -t N.NNN support (+110 bytes) (FEATURE_SH_READ_FRAC) [Y/n/?] (NEW) y
Use SHISTFILESIZE (FEATURE_SH_HISTFILESIZE) [Y/n/?] (NEW) y
Embed scripts in the binary (FEATURE_SH_EMBEDDED_SCRIPTS) [Y/n/?] (NEW) y
*
* System Logging Utilities
*
klogd (6.2 kb) (KLOGD) [Y/n/?] (NEW) y
*
* klogd should not be used together with syslog to kernel printk buffer
*
Use the klogctl() interface (FEATURE_KLOGD_KLOGCTL) [Y/n/?] (NEW) y
logger (6.5 kb) (LOGGER) [Y/n/?] (NEW) y
logread (5 kb) (LOGREAD) [Y/n/?] (NEW) y
Double buffering (FEATURE_LOGREAD_REDUCED_LOCKING) [Y/n/?] (NEW) y
syslogd (14 kb) (SYSLOGD) [Y/n/?] (NEW) y
Rotate message files (FEATURE_ROTATE_LOGFILE) [Y/n/?] (NEW) y
Remote Log support (FEATURE_REMOTE_LOG) [Y/n/?] (NEW) y
Support -D (drop dups) option (FEATURE_SYSLOGD_DUP) [Y/n/?] (NEW) y
Support syslog.conf (FEATURE_SYSLOGD_CFG) [Y/n/?] (NEW) y
Include milliseconds in timestamps (FEATURE_SYSLOGD_PRECISE_TIMESTAMPS) [N/y/?] (NEW) n
Read buffer size in bytes (FEATURE_SYSLOGD_READ_BUFFER_SIZE) [256] (NEW) 256
Circular Buffer support (FEATURE_IPC_SYSLOG) [Y/n/?] (NEW) y
Circular buffer size in kbytes (minimum 4KB) (FEATURE_IPC_SYSLOG_BUFFER_SIZE) [16] (NEW) 16
Linux kernel printk buffer support (FEATURE_KMSG_SYSLOG) [Y/n/?] (NEW) y
```

2. Изменение конфигурационного файла:

Были изменены такие параметры:

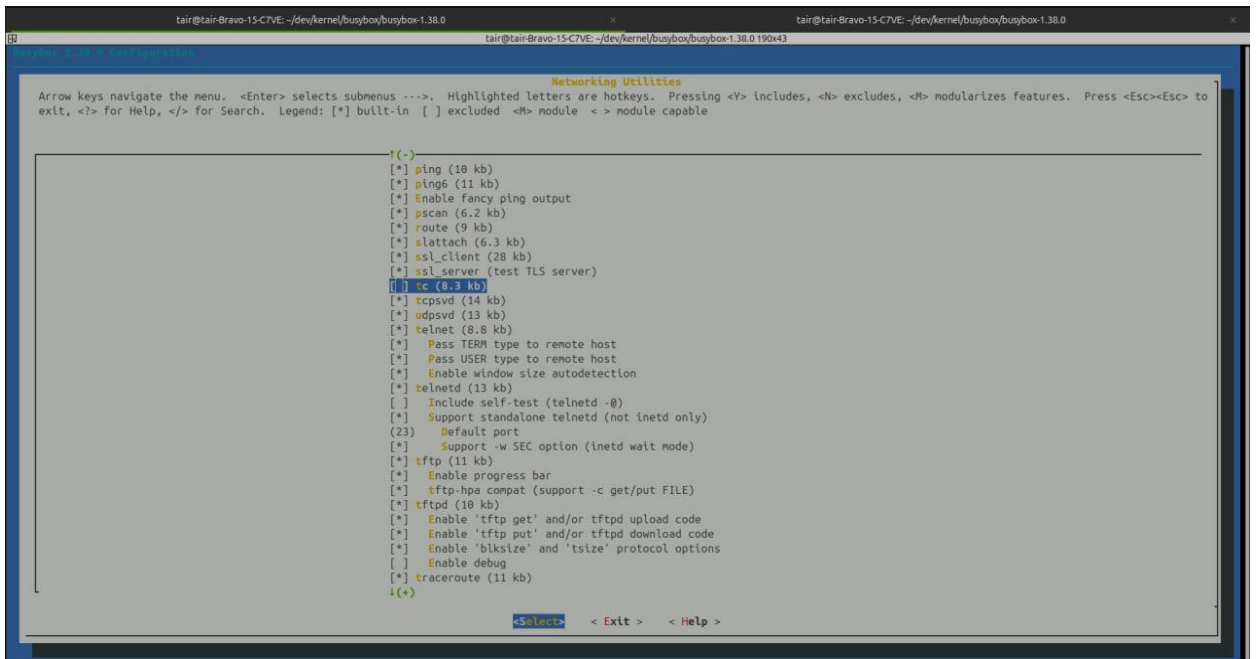
- Build static binary
- tc (отключил, из-за того, что компилятор выдавал ошибки)
- добавил префикс компиляции

ARCH=arm make menuconfig

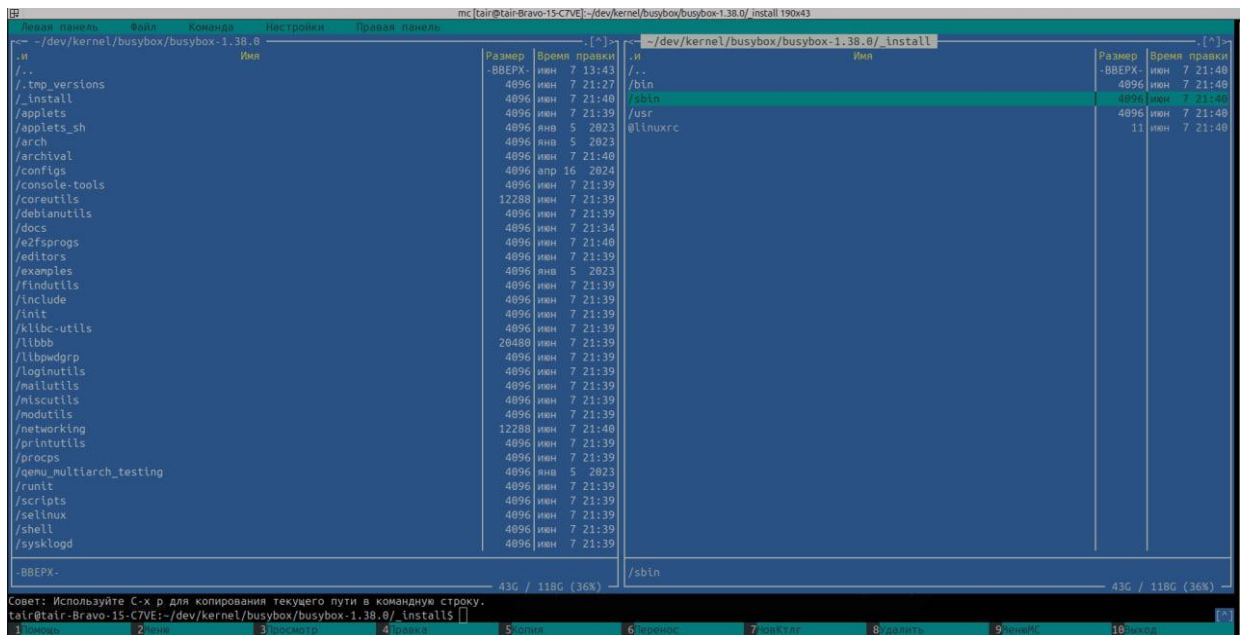
```
tair@tair-Bravo-15-C7VE: ~/dev/kernel/busybox/busybox-1.38.0
BusyBox 1.38.0 Configuration

Arrow keys navigate the menu. <Enter> selects submenus ---. Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to
exit, <?> for Help, </> for Search. Legend: [*] built-in [ ] excluded <M> module <-> module capable

Settings
[*] Support writing pidfiles
(/var/run) B Directory for pidfiles
[*] Include busybox applet
[*] Support --show SCRIPT
[*] Support --install [-s] to install applet links at runtime
[*] Support --version
[ ] Don't use /usr
[*] Drop SUID state for most applets
[*] Enable SUID configuration via /etc/busybox.conf
[*] Suppress warning message if /etc/busybox.conf is not readable
[ ] exec prefers applets
(/proc/self/exe) P Path to busybox executable
[ ] Support NSA Security Enhanced Linux
[ ] Clean up all memory before exiting (usually not needed)
[*] Support LOG_INFO level syslog messages
--- Build Options
[*] Build static binary (no shared libs)
[ ] Force NOMMU build
(arm-linux-gnueabihf-) C Cross compiler prefix
() Path to sysroot
() Additional CFLAGS
() Additional LDFLAGS
() Additional LDLIBS
[ ] Avoid using GCC-specific code constructs
[*] Use -npreferrred-stack-boundary=2 on i386 arch
[*] Use -static-libgcc
--- Installation Options ("make install" behavior)
What kind of applet links to install (as soft-links) ---
(./_install) Destination path for 'make install'
+(*+)
<Select> < Exit > < Help >
```



3. Сборка файловой системы: make install



[illegible]